

Dome Inspection and Metrology

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1 Context

In the first week of December 2025, the neoprene seal between the building and the dome was damaged. It appears to have been torn at the point that it is fixed by steel strips to the interface plates attached to the building. It fell between the dome and building on the SW side and prevented movement. On-site staff removed the damaged seal between azimuths of about 150 and 330 degrees and fixed the ends of the undamaged seal so that the dome could turn. See Figure 1.

2 Are the Supports Damaged?

During December 2025, I inspected the radial and vertical supports of the dome. I found no evidence that any of them were damaged, bent, or had moved.

3 Is the Dome Centered?

During December 2025, with the help of Santiago Paez, I performed metrology on the dome.

All azimuths are measured from north through east. We assume the northernmost radial support is at an azimuth of zero. This is probably not completely correct, but it is close, and this support provides a convenient reference.

3.1 With Respect to the Building

We measured the distance D between the outer radius of the beam on the building and the inner radius of the dome at the center of 14 of the 16 sizes to the building. We were not able to measure this at two positions over the door. See Figure 2. The dome was positioned with the slit to the north. The results are shown in Figure 3 along with a sinusoidal model (fit by eye). We repeated the measurement twice ($D1$ and $D2$) to confirm that the point at 292.5 degrees, is correct. The figure suggests that the dome is displaced with respect to the building by about 15 mm, being furthest from the building at an azimuth of about 300 degrees (WNW) and closest at about 120 degrees (ESE). At its

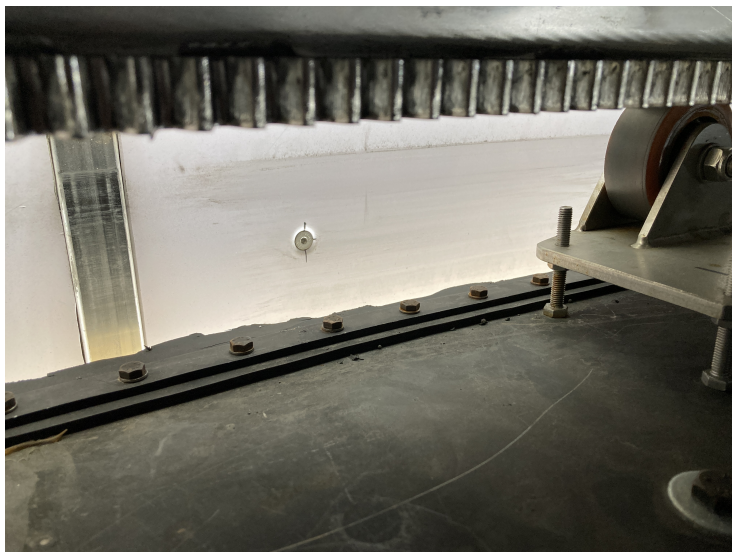


Figure 1: The damaged seal.

widest point, the gap is almost 50 mm, but at its narrowest point it is about 15 mm.

Figure 4 shows the impact of displacing the dome by 15 mm at an azimuth of 300 degrees, assuming the dome moves rigidly. Most of the points are more between about 25 and 35 mm. However, the discrepant point at 292.5 degrees is at about 15 mm. We are not sure whether this point is (a) a discrepant point in the building radius (i.e., the beam extends more than the other points) or (b) a discrepant point in the dome radius (i.e., the dome does not extend as far as at other points) or both.

3.2 With Respect to the Interface Plate

We measured the distance G from the inner radius of the interface plates to the bottom of the teeth in the drive track. We did this with the dome at azimuths of 0, 90, 180, and 270 degrees. The first couple of positions showed high variability, but we checked them and it is real.



Figure 2: Measuring the distance D between the outer radius of the building beam and inner radius of the dome.

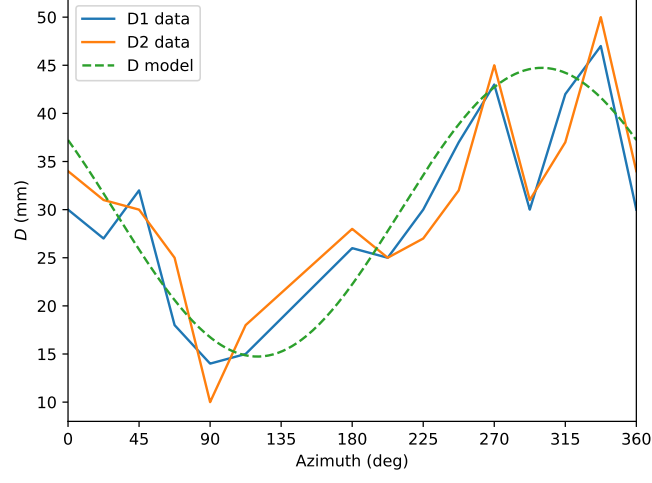


Figure 3: The distance D between the outer radius of the building beam and inner radius of the dome as a function of azimuth. The plot shows two sets of measurements ($D1$ and $D2$) and an approximate sinusoidal model with a half-amplitude of 15 mm.

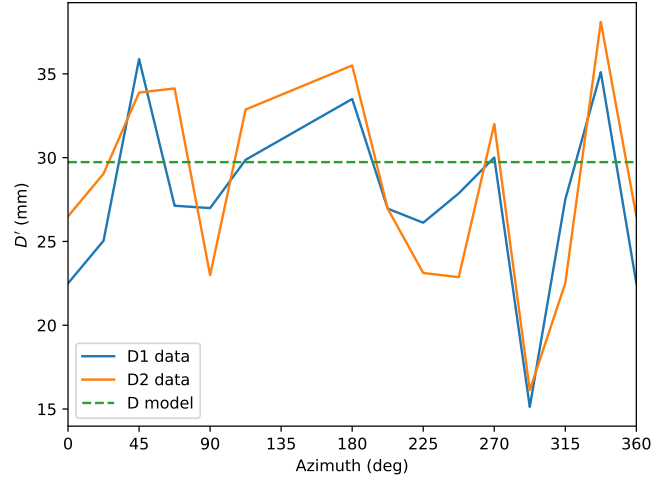


Figure 4: The estimated distance D' between the outer radius of the building beam and inner radius of the dome as a function of azimuth after displacing the dome by 15 mm at an azimuth of 300 degrees. The model assumes the dome moves rigidly.

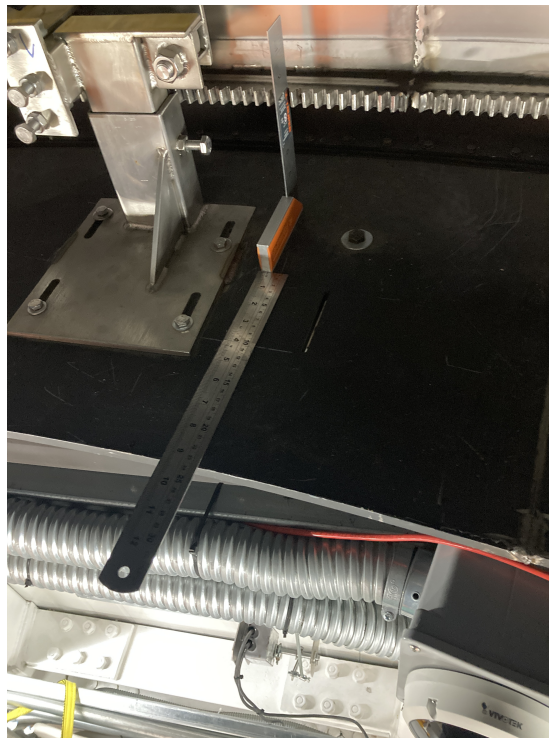


Figure 5: Measuring the distance G from the inner radius of the interface plates to the bottom of the teeth in the drive track.